

Comparative Effectiveness of Oral Monotherapy Versus Dual Therapy on HbA1c Levels Among Adults with Type 2 Diabetes Mellitus and Hyperglycemia

Using 2017 - 2020 NHANES Data

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Background

Type 2 diabetes mellitus (T2DM) is one of the most prevalent chronic diseases in the United States, affecting approximately 12% of the population, with over one-quarter of cases remaining undiagnosed. In addition, a substantially larger population has prediabetes, placing them at high risk for progression to overt diabetes. These trends highlight the significant and growing public health burden of T2DM.

Effective glycemic control is essential for reducing diabetes-related complications. Pharmacologic therapy plays a central role in management, with metformin recommended as first-line treatment due to its ability to reduce hepatic glucose production and improve insulin sensitivity. Sulfonylureas, which stimulate insulin secretion, are also widely used because of their strong glucose-lowering effects and affordability. Prior studies suggest that dual therapy combining metformin and sulfonylureas may achieve greater reductions in HbA1c compared to monotherapy.

The objective of this study is to evaluate the comparative effectiveness of oral monotherapy versus dual therapy (metformin plus sulfonylurea) in reducing HbA1c levels among U.S. adults.

The primary research question is: among adults with T2DM or hyperglycemia in NHANES, does dual therapy result in different HbA1c levels compared to monotherapy after adjusting for demographic and clinical factors?

Methods

This retrospective cross-sectional study used data from the National Health and Nutrition Examination Survey (NHANES) 2017 - March 2020 pre-pandemic cycle. NHANES is a nationally representative survey of the U.S. population that combines interview and examination data.

The study population included adults aged 21 years or older who reported a prior diagnosis of diabetes or prediabetes by a healthcare professional, completed both the interview and examination components, and reported use of either metformin or a

sulfonylurea within the past 365 days. Participants with complete data on exposure (oral monotherapy vs dual therapy) and outcome (HbA1c level) were included, resulting in a final analytic sample of 737 individuals, of whom 177 received dual therapy.

The exposure was defined as treatment with oral dual therapy (metformin plus sulfonylurea) versus monotherapy with either agent alone. The primary outcome was HbA1c level, measured during the NHANES examination.

To address potential confounding, propensity scores were estimated using logistic regression, incorporating the following covariates: age, sex, race/ethnicity, body mass index, systolic blood pressure, alcohol consumption, healthcare utilization in the past 12 months, and primary insurance coverage.

Participants receiving dual therapy were matched 1:1 to those receiving monotherapy using greedy nearest-neighbor matching without replacement. Sensitivity analyses were conducted to assess the potential impact of unmeasured confounding. In addition, inverse probability weighting targeting the average treatment effect on the treated (ATT) was applied. A doubly robust approach was used to further adjust for residual confounding. After applying propensity score methods, differences in HbA1c levels between treatment groups were estimated using weighted linear regression models.

Results

The results consistently show that, compared with monotherapy, dual therapy is associated with higher HbA1c levels among patients with T2DM or hyperglycemia. In the unadjusted analysis, patients receiving dual therapy had HbA1c levels that were 0.578% higher than those receiving monotherapy (95% CI: 0.319, 0.837). This association remained consistent after adjustment using both 1:1 greedy matching and ATT weighting, with estimated differences of 0.518 (95% CI: 0.192, 0.843) and 0.598 (95% CI: 0.325, 0.871), respectively.

In terms of covariate balance, the unadjusted data exhibited substantial imbalance, with standardized differences ranging from -19 to 18.2. This imbalance was reduced after matching (-6.8 to 5.9) and also minimized by ATT weighting (-3.2 to 3.5). Overall, as covariate balance improved, the estimated treatment effects became more reliable, suggesting that proper adjustment for confounding is essential in estimating the association between treatment and HbA1c levels.

The estimated differences in HbA1c were very similar across all three analytic

approaches, indicating that adjustment for measured confounding did not substantially alter the results. In particular, the ATT weighting analysis produced an estimate that was nearly identical to the unadjusted result, suggesting that weighting did not meaningfully change the composition of the analytic population. 1:1 greedy propensity score matching yielded a slightly smaller estimated difference. This may reflect the fact that matching restricts the analysis to a more comparable subset of patients by discarding unmatched individuals, whereas weighting retains a broader population and targets the average treatment effect among the treated. Additionally, sensitivity analysis indicated that the matched results were somewhat sensitive to unmeasured confounding, suggesting that the observed association should be interpreted with caution.

Conclusions

Across the three analytic approaches, the estimated treatment effects were consistent in direction. The unadjusted analysis showed that subjects receiving dual therapy had an HbA1c level that was on average 0.578% higher than those receiving monotherapy (95% CI: 0.319 to 0.837). After adjustment using 1:1 greedy matching, the estimated difference increased to 0.518% (95% CI: 0.192 to 0.843). The ATT weighting approach yielded an intermediate estimate, with dual therapy associated with a 0.598% higher HbA1c level compared to monotherapy (95% CI: 0.325, 0.871).

One possible explanation for the higher HbA1c levels observed among subjects receiving dual therapy is confounding by indication. Patients are typically prescribed combination therapy when their glycemic control is inadequate under monotherapy or when their disease is more severe. Individuals receiving both metformin and sulfonylureas may already have higher baseline HbA1c levels or more advanced diabetes compared to those receiving a single agent. Although propensity score methods were used to adjust for observed covariates, residual confounding may still exist due to unmeasured factors such as duration of diabetes, medication adherence, or lifestyle behaviors that are not fully captured in the data.